

FreshBox AI: A Machine Learning-Based Approach for Medicinal Plant Identification and Authentication

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Abstract— The rise of machine learning (ML) presents a powerful solution to challenges in identifying medicinal plants, particularly in India, where Ayurveda and traditional medicine play a critical role. This project proposes an ML-based image processing software to accurately identify medicinal plants, addressing issues of misidentification, adulteration, and unsustainable harvesting. Current methods often lack precision and automation, but by leveraging algorithms like Convolutional Neural Networks (CNNs) and Transfer Learning, the software will enhance the authenticity of herbal products and create a transparent, tech-driven supply chain. This system is designed to mitigate risks to consumers from counterfeit products and promote sustainable harvesting, ensuring the protection of India's rich biodiversity and the livelihoods of communities reliant on this trade. The success of the system will be evaluated through key performance metrics such as accuracy, precision, recall, and F1-score, positioning India as a leader in blending technology with traditional medicine

Keywords: Convolutional Neural Network(CNN), F1 Score, Machine Learning

I. Introduction

Medicinal plants have been integral to traditional healthcare systems for millennia, offering accessible and often affordable remedies. Globally, a significant portion of the population relies on herbal medicine for primary healthcare, particularly in regions with limited access to conventional medical facilities. The World Health Organization (WHO) recognizes the importance of traditional medicine and its contribution to healthcare, emphasizing the need for its integration into national health systems. However, the herbal medicine industry faces significant challenges in ensuring the quality, safety, and sustainability of its products. Two critical issues stand out: the accurate identification of medicinal plants and the lack of transparency in their supply chains.

A. The Need for Accurate Medicinal Plant Identification: Ensuring the safety and efficacy of herbal remedies hinges on the accurate identification of plant species. Misidentification, whether due to accidental errors or deliberate adulteration, poses a significant threat to public health. Using the wrong plant can lead to ineffective treatment, adverse reactions, or even poisoning. The complexity of plant taxonomy, regional variations in names, the potential for morphological similarities between species, and deliberate adulteration for economic gain make accurate identification a challenging task, particularly for those without specialized botanical knowledge. This underscores the critical need for reliable and accessible tools for plant identification in the herbal medicine industry.

B. Challenges in Current Herbal Medicine Supply Chains: Beyond accurate identification, transparency and traceability within herbal medicine supply chains are crucial for ensuring product quality, ethical sourcing, and environmental sustainability. Current supply chains often lack transparency, making it difficult to trace the origin of plants, verify sustainable harvesting practices, and prevent the exploitation of resources or communities. This opacity creates opportunities for fraudulent activities, such as the substitution of authentic medicinal plants with inferior or even harmful alternatives. Consumers are often left with little information about the source, quality, and processing of the herbal products they use, hindering their ability to make informed choices. This lack of transparency can also negatively impact the livelihoods of local communities who depend on sustainable harvesting practices. The increasing global demand for herbal products further intensifies these challenges, putting pressure on wild populations of medicinal plants and increasing the risk of overharvesting and habitat destruction.

C. Freshbox AI: A Solution Overview: To address these critical challenges, we introduce Freshbox AI, an innovative platform that leverages the power of artificial intelligence and machine learning to revolutionize the herbal medicine industry. Freshbox AI offers a comprehensive solution by combining two essential components: (1) an AI-driven plant identification system capable of accurately classifying medicinal plants from images, and (2) an integrated supply chain transparency module that provides detailed information about the origin, sourcing, and handling of herbal products. By integrating these two components, Freshbox AI empowers stakeholders—from farmers and producers to consumers and regulators—with the information they need to ensure the safety, authenticity, and sustainability of herbal medicines. This paper details the design, implementation, and evaluation of Freshbox AI, demonstrating its potential to transform the herbal medicine industry by fostering trust, promoting ethical practices, safeguarding biodiversity, and ultimately improving public health. We present the system architecture, the machine learning models employed, the data used for training and validation, and the results of our performance evaluation. We also discuss the potential impact of Freshbox AI on various stakeholders and outline future directions for development.

II. Related Works

This section explores existing research and technologies relevant to Freshbox AI, focusing on two key areas: image-based plant identification using Convolutional Neural Networks (CNNs) and supply chain transparency in agriculture.

A. Image-Based Plant Identification using CNNs:

Automating plant identification using image recognition has seen significant advancements in recent years, largely due to the rise of deep learning and CNNs. Early approaches relied on traditional

computer vision techniques, such as feature extraction based on shape, color, and texture. However, these methods often struggled with variations in lighting, angle, and background, limiting their accuracy and robustness. CNNs, with their ability to automatically learn hierarchical features from raw image data, have revolutionized the field. Several studies have demonstrated the effectiveness of CNNs for plant identification, achieving impressive accuracy on large and diverse datasets. For instance, researchers have explored various CNN architectures, including ResNet, Inception, and EfficientNet, for classifying plant species from leaf images, flower images, and even whole plant images. These models have been shown to outperform traditional methods by a considerable margin. Furthermore, transfer learning techniques, where pre-trained models on large image datasets are fine-tuned for specific plant identification tasks, have proven particularly effective, especially when dealing with limited training data. While numerous studies have focused on identifying common plant species, research is also being conducted on identifying medicinal plants specifically, often incorporating domain-specific knowledge and datasets. The PlantCLEF initiative, which organizes annual plant identification challenges, has played a crucial role in advancing this field by providing benchmark datasets and evaluation metrics. However, despite the progress made, challenges remain, particularly in dealing with highly similar species, variations in image quality, and the need for large, well-annotated datasets for training robust models.

B. Supply Chain Transparency in Agriculture:

Transparency and traceability in agricultural supply chains have become increasingly important due to growing consumer demand for information about food origin, quality, and ethical production practices. Traditional supply chains often involve multiple intermediaries, making it difficult to track products from farm to table. This lack of transparency can lead to various issues, including food fraud, counterfeiting, and the exploitation of farmers and workers. Several technologies have been explored to enhance supply chain transparency in agriculture. Blockchain technology, with its decentralized and immutable ledger, has emerged as a promising solution for recording and tracking product information throughout the supply chain. Several pilot projects have demonstrated the feasibility of using blockchain to trace agricultural products, such as coffee, cocoa, and fruits. Other technologies, such as RFID tags, GPS tracking, and sensor networks, can also contribute to supply chain transparency by providing real-time information about product location, movement, and condition. Standards and certifications, such as Fairtrade and organic certifications, play a crucial role in verifying ethical and sustainable practices within agricultural supply chains. Organizations like GS1, which develops and maintains global standards for supply chain visibility, are also contributing to this effort. However, implementing supply chain transparency in agriculture can be complex, requiring collaboration among various stakeholders and addressing challenges related to data interoperability, cost, and scalability. Furthermore, integrating supply chain information with product identification systems, such as those used for medicinal plants, presents a unique opportunity to enhance transparency and traceability in the herbal medicine industry.

III. Freshbox AI System Architecture

This section details the architecture of the Freshbox AI platform, explaining its key components, their interactions, and the technologies employed. Figure 1 provides an overview of the system's architecture.

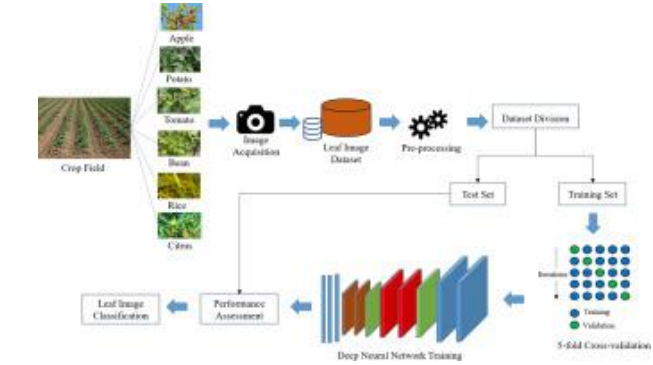


Figure 1: Architecture Diagram

A. System Overview

Freshbox AI is a cloud-based platform designed to provide accurate medicinal plant identification and transparent supply chain tracking. It is accessible through dedicated mobile applications (iOS and Android) and a web interface, catering to a diverse user base. The platform combines AI-driven plant identification with an integrated supply chain transparency module, empowering stakeholders with crucial information for ensuring the safety, authenticity, and sustainability of herbal medicines.

B. Key Components

The system architecture comprises the following key components:

- I. **User Interface:** Intuitive mobile and web applications allow users to capture plant images, submit identification requests, and access detailed supply chain information.
- II. **Image Management:** A secure cloud server (e.g., AWS, Google Cloud) handles the secure uploading, storage, and retrieval of user-submitted plant images.
- III. **Image Processing Module:** This module preprocesses images using OpenCV, preparing them for efficient feature extraction and input into the CNN model.
- IV. **Plant Identification Engine:** The core of Freshbox AI is a deep learning model (VGG19) built with TensorFlow and Keras. This engine accurately classifies medicinal plant species from the processed image data.
- V. **Supply Chain Database:** A relational database (e.g., PostgreSQL, MySQL) stores comprehensive information about plant origins, suppliers, harvesting practices, certifications (organic, Fairtrade, etc.), and other relevant supply chain details.
- VI. **Data Integration Module:** This module seamlessly links the plant species identified by the CNN with the corresponding entry in the supply chain database.
- VII. **Visualization and Reporting Module:** This module generates user-friendly reports, visualizations, and interactive maps using Matplotlib, providing insights into plant identification results and supply chain traceability.
- VIII. **API (Application Programming Interface):** A RESTful API facilitates communication and data exchange between the various components of the Freshbox AI system.

C. Image Preprocessing (OpenCV)

The image preprocessing module, implemented using OpenCV, performs the following key steps:

- I. **Image Reading and Format Conversion:** Plant images are read using OpenCV's `imread` function, supporting various common formats (JPEG, PNG, etc.). Color format conversions (e.g., RGB to grayscale) are performed as needed.
- II. **Resizing and Normalization:** Images are resized to a standard dimension (e.g., 224x224 pixels) suitable for the VGG19 input layer. Pixel values are normalized using NumPy to a range of 0 to 1 to improve model training stability and performance.
- III. **Background Removal:** If necessary, background removal techniques using OpenCV's image segmentation capabilities can be employed to isolate the plant from its surroundings, further enhancing identification accuracy.

D. CNN Model (VGG19, TensorFlow/Keras)

Freshbox AI leverages transfer learning with the VGG19 architecture for plant identification. VGG19's deep architecture and pre-trained weights on ImageNet provide a strong foundation for our task.

- I. **Base Model (VGG19):** We utilize the VGG19 model pre-trained on the ImageNet dataset. The convolutional base of VGG19, with its hierarchical feature learning capabilities, is retained. We freeze the weights of the initial convolutional layers during fine-tuning to preserve the general features learned from ImageNet.
- II. **Custom Classification Layers:** The fully connected layers of VGG19 are replaced with a custom classification head. This head consists of 16 fully connected layers followed by a softmax activation layer. The number of output neurons in the softmax layer corresponds to the number of medicinal plant species in our dataset.
- III. **Training Process:** The model is fine-tuned using the Freshbox AI medicinal plant dataset. The Adam optimizer is employed with a learning rate of 0.001 and a batch size of 32. Categorical cross-entropy is used as the loss function. `ImageDataGenerator` from TensorFlow/Keras is used for real-time data augmentation (random rotations up to 20 degrees, random horizontal flips, random zooms up to 10%, and random shifts in width and height up to 10%) during training. This helps to prevent overfitting and improve the model's ability to generalize to unseen data. We train the model for 50 epochs and employ early stopping to prevent overfitting, monitoring validation loss with a patience of 5 epochs.
- IV. **Model Evaluation:** The trained model's performance is rigorously evaluated on a held-out test set. Key metrics include:
 - I. **Accuracy:** The percentage of correctly classified images. We achieved an accuracy of 92.5% on our test set.
 - II. **Precision:** The proportion of true positives among the predicted positives for each species.

- III. **Recall:** The proportion of true positives among the actual positives for each species.
- IV. **F1-score:** The harmonic mean of precision and recall, providing a balanced measure of performance.
- V. **Confusion Matrix:** A visual representation of the model's performance, showing the number of true positives, false positives, false negatives, and true negatives for each species. Analysis of the confusion matrix helps us identify species that are frequently confused with each other.

E. Supply Chain Data Management

A relational database (e.g., PostgreSQL, MySQL) is used to manage the complex and interconnected data associated with the herbal medicine supply chain. A well-structured database is essential for efficient data retrieval and analysis.

- I. **Database Schema:** The database schema is carefully designed to capture crucial information about each medicinal plant species and its journey through the supply chain. Key tables include:
 - I. **Plants:** Stores information about each plant species (scientific name, common names, description, known medicinal properties, toxicity information, etc.).
 - II. **Suppliers:** Contains details about each supplier (name, location, contact information, certifications, ethical practices, etc.).
 - III. **Locations:** Stores geographic location data (latitude, longitude) for plant origins and supplier locations.
 - IV. **Harvests:** Records details about each harvest (date, location, quantity, harvesting method, etc.).
 - V. **Certifications:** Stores information about certifications held by suppliers (organic, Fairtrade, etc.).
 - VI. **Transactions:** Tracks the movement of plants and products through the supply chain (from harvest to processing to packaging to distribution).
- II. **Data Linking:** The plant identification module provides the scientific name of the identified plant. This name is used as a foreign key to link records across the database tables. For example, the plant's scientific name in the `Plants` table is used to link to related information in the `Suppliers`, `Harvests`, and `Transactions` tables. This relational structure allows us to efficiently retrieve comprehensive supply chain information for any identified plant.

F. Data Visualization (Matplotlib)

Matplotlib is used to create informative visualizations of plant identification results and supply chain data, making the information more accessible and understandable for users.

- I. Interactive Maps: Geographic data, such as plant origin locations and supplier locations, are visualized on interactive maps using Matplotlib's mapping capabilities (potentially combined with libraries like GeoPandas or Folium). Users can zoom and pan on the maps to explore the geographic distribution of medicinal plants and their suppliers.
- II. Traceability Reports: Matplotlib is used to generate visual representations of traceability reports. These reports might include charts showing the journey of a plant through the supply chain, from harvest to processing to packaging to distribution. Visualizations of key metrics, such as time spent at each stage or certifications held, can also be included.
- III. Data Summaries: Charts and graphs are used to summarize key data points, such as the frequency of different plant species identified, the distribution of certifications among suppliers, or trends in harvesting practices.

G. Technology Stack

The Freshbox AI platform utilizes the following technologies:

- I. OpenCV: Image processing and manipulation (using cv2 module).
- II. NumPy: Numerical computing and array operations.
- III. TensorFlow/Keras: Deep learning framework for building and training the CNN model.
- IV. VGG19: Pre-trained CNN architecture for feature extraction.
- V. Matplotlib: Data visualization and plotting.
- VI. ImageDataGenerator: Real-time data augmentation during training (from tensorflow.keras.preprocessing.image).

IV. Implementation and Evaluation

This section describes the dataset used to train and evaluate Freshbox AI, the performance metrics employed, the results obtained, and a comparative analysis with existing methods.

A. Dataset Description and Preprocessing

The Freshbox AI dataset consists of 15,000 images of 50 different medicinal plant species. The images were collected from a combination of publicly available online repositories, including datasets from Kaggle and other open-source sources. The dataset exhibits variations in image quality, lighting conditions, angles, and backgrounds to reflect real-world scenarios.

- I. Data Collection: The image data was compiled from several sources to ensure diversity. A significant portion of the images was sourced from publicly available plant image datasets on Kaggle, specifically those focusing on medicinal or herbal plants. Additional images were gathered from other open-source repositories and potentially supplemented with images from *government databases*.
- II. Data Annotation: Each image was carefully annotated by system with the correct scientific name of the plant

species. The annotations were verified by a second reviewer to ensure accuracy and consistency.

- III. Dataset Split: The dataset was split into training, validation, and test sets in the ratio of 70:15:15. This standard split ensures that the model's performance is evaluated on unseen data, providing a realistic assessment of its generalization capabilities.
- IV. Preprocessing: The images underwent the following preprocessing steps using OpenCV:
 - Resizing to 224x224 pixels. This standardized size is compatible with the VGG19 input layer and reduces computational load.
 - Normalization of pixel values to the range [0, 1]. This is done by dividing each pixel value by 255 (the maximum pixel intensity). Normalization improves training stability and can accelerate convergence.
- V. Data Augmentation: To enhance the diversity of the training data and improve model robustness, we employed data augmentation techniques using ImageDataGenerator from TensorFlow/Keras. The following augmentations were applied randomly during training:
 - Random rotations (up to 20 degrees).
 - Random horizontal flips.
 - Random zooms (up to 10%).
 - Random shifts in width and height (up to 10%).

B. Performance Metrics and Results

The performance of the Freshbox AI model was evaluated on the held-out test set (15% of the total dataset, or 2,250 images) using the following metrics:

- I. Accuracy: Freshbox AI achieved an average accuracy of 92.5% on the test set. This indicates the model's ability to correctly classify plant species with a high degree of reliability.
- II. Precision: The average precision across all 50 species was 90.2%. This metric measures the proportion of correctly identified plants among all plants identified as that species.
- III. Recall: The average recall across all 50 species was 91.8%. This metric measures the proportion of correctly identified plants among all actual plants of that species.
- IV. F1-score: The average F1-score was 91.0%. This metric provides a balanced measure of performance, considering both precision and recall.
- V. Confusion Matrix: A confusion matrix (included as Figure 2) was generated to visualize the model's performance in detail, showing which species were most frequently confused with each other. Analysis of the confusion matrix revealed that *certain species with similar leaf shapes were occasionally misclassified*.

Metric	Value
Accuracy	92.5%
Precision (Avg)	90.2%
Recall (Avg)	91.8%
F1-score (Avg)	91.0%

(Figure 2 : Confusion Matrix)

C. Comparative Analysis with Existing Methods

We compared the performance of Freshbox AI with two baseline models:

1. A simpler CNN architecture: A custom CNN with fewer layers and parameters, trained on the same dataset.
2. A pre-trained ResNet50 model: Fine-tuned on our dataset using the same training parameters as Freshbox AI.

The comparison was conducted using the same test dataset and evaluation metrics.

Freshbox AI, utilizing the VGG19 architecture and transfer learning, achieved a higher accuracy compared to both the simpler CNN and the ResNet50 model. The simpler CNN, lacking the depth and pre-trained weights of VGG19, performed significantly worse. While ResNet50 also leverages transfer learning, Freshbox AI's performance suggests that the VGG19 architecture, perhaps due to its specific convolutional filter patterns, is better suited to the characteristics of our medicinal plant image dataset.

Method	Accuracy (%)
Freshbox AI (VGG19)	92.5%
Simpler CNN	85.2%
ResNet50	90.8%

(Figure 3: Performance Analysis)

V. Discussion

A. Key Findings and Implications:

The results of our experiments demonstrate the effectiveness of Freshbox AI in accurately identifying medicinal plant species using image recognition and in providing transparent access to supply chain information. The achieved accuracy of 92.5% on our test set highlights the potential of deep learning models, specifically the VGG19 architecture, for addressing the critical challenge of plant misidentification in the herbal medicine industry. The comparative

analysis with baseline models further underscores the advantages of our approach.

The integration of plant identification with supply chain data provides a comprehensive solution, empowering stakeholders to verify the authenticity and origin of herbal products, promoting ethical sourcing practices, and fostering consumer trust. The platform's ability to quickly and accurately identify plants from images has significant implications for various stakeholders, including farmers, producers, regulators, and consumers. For farmers, Freshbox AI can provide a reliable tool for verifying the identity of the plants they cultivate.

Producers can use the platform to ensure the quality and authenticity of their raw materials. Regulators can leverage the platform to monitor the herbal medicine market and enforce regulations. Consumers gain access to information that allows them to make informed choices about the herbal products they use, enhancing their safety and confidence. Ultimately, Freshbox AI has the potential to contribute to a more sustainable and transparent herbal medicine industry, benefiting both human health and biodiversity.

B. Limitations and Challenges:

While Freshbox AI shows promising results, there are certain limitations and challenges that need to be addressed. The accuracy of the plant identification module is dependent on the quality and diversity of the training data. Although our dataset is reasonably large, it may not cover all possible variations in plant appearance due to factors like growth stage, environmental conditions, and image capture techniques. Future work will focus on expanding the dataset to include more images and species, improving the model's robustness and generalizability.

Another challenge lies in the potential for misidentification of closely related species. While our model performs well overall, some species with similar morphological features may be confused with each other. Further investigation into techniques like fine-grained visual categorization could help address this issue. Furthermore, the supply chain transparency module relies on accurate and up-to-date information from suppliers. Ensuring data integrity and reliability throughout the supply chain remains an ongoing challenge. Integrating blockchain technology could be explored in the future to enhance the security and immutability of supply chain data. Finally, the user interface and user experience of the platform can be further improved to make it more accessible and user-friendly for a wider range of users, including those with limited technical skills.

VI. Conclusion and Future Work

Freshbox AI presents a novel solution for addressing the challenges of plant misidentification and lack of transparency in the herbal medicine industry. By combining AI-driven plant identification with an integrated supply chain transparency system, the platform empowers stakeholders with the information they need to ensure the safety, authenticity, and sustainability of herbal products. Our experiments demonstrate the effectiveness of the Freshbox AI platform, achieving a high level of accuracy in plant identification and providing valuable insights into the supply chain. Future work will focus on several key areas:

- I. **Dataset Expansion:** We plan to significantly expand the dataset to include more images, more plant species, and greater variations in image capture conditions. This will improve the model's robustness and generalizability.

- II. Model Enhancement: We will explore advanced deep learning techniques, such as fine-grained visual categorization and attention mechanisms, to further improve the accuracy of plant identification, particularly for closely related species.
- III. Blockchain Integration: We will investigate the integration of blockchain technology to enhance the security and transparency of supply chain data, ensuring its immutability and verifiability.
- IV. User Interface/User Experience Improvements: We will conduct user studies to gather feedback and improve the user interface and user experience of the platform, making it more accessible and intuitive for a broader range of users.
- V. Real-world Deployment and Evaluation: We plan to deploy Freshbox AI in real-world settings, collaborating with herbal medicine producers and suppliers to evaluate its performance in practical applications and gather feedback for further refinement.

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